

ETL process:

```
import pandas as pd

data = pd.read_csv('sales.csv')

data.fillna(0, inplace=True)

data['Total'] = data['Quantity'] * data['Price']

data.to_csv('warehouse_sales.csv', index=False)
```

Analyzing Multidimensional Data:

```
import pandas as pd

data = {

    "Time": ["2024-Q1", "2024-Q1", "2024-Q1", "2024-Q2", "2024-Q2", "2024-Q2"],

    "Region": ["North", "South", "East", "North", "South", "East"],

    "Product": ["Laptop", "Laptop", "Laptop", "Mobile", "Mobile", "Mobile"],

    "Sales": [20000, 15000, 18000, 22000, 17000, 19000]

}

// Data cube creation

df = pd.DataFrame(data)

olap_cube = pd.pivot_table(

    df,

    values="Sales",

    index=["Time", "Region"],

    columns="Product",

    aggfunc="sum",

    fill_value=0
```

```
)
```

```
# Sales of Laptop only
```

```
laptop_sales = olap_cube["Laptop"]
```

```
print(laptop_sales)
```

```
# Sales in 2024-Q2
```

```
q2_sales = olap_cube.loc["2024-Q2"]
```

```
print(q2_sales)
```

```
# Total sales by Time
```

```
rollup_time = df.groupby("Time")["Sales"].sum()
```

```
print(rollup_time)
```

```
# Region-wise sales for each time period
```

```
drilldown_region = df.groupby(["Time", "Region"])["Sales"].sum()
```

```
print(drilldown_region)
```

```
# Region vs Product analysis
```

```
pivot_view = pd.pivot_table(
```

```
    df,
```

```
    values="Sales",
```

```
    index="Region",
```

```
    columns="Product",
```

```
    aggfunc="sum"
```

```
)
```

```
print(pivot_view)
```